

Dr. Elena Rostova

Postdoctoral Research Fellow | Distributed Systems & Machine Learning

Department of Computer Science, Stanford University | elena.rostova@cs.stanford.edu | (650) 498-1240

EDUCATION

Ph.D. in Computer Science

2018 - 2023

Massachusetts Institute of Technology (MIT), Cambridge, MA

Dissertation: Scalable Fault-Tolerant Consensus for Distributed Systems

Advisor: Prof. David K. Gifford

B.S. in Computer Science & Applied Mathematics, Summa Cum Laude

2014 - 2018

Carnegie Mellon University, Pittsburgh, PA

RESEARCH EXPERIENCE

Postdoctoral Research Fellow

2023 - Present

Stanford University, Stanford, CA

? Investigating asynchronous replication protocols for geo-distributed deep learning pipelines.

? Published 2 primary papers in top-tier conferences (OSDI, SOSP). Mentored 3 graduate students.

Graduate Research Assistant

2018 - 2023

Computer Science and Artificial Intelligence Laboratory (CSAIL), MIT

? Designed and benchmarked a novel Byzantine fault-tolerant consensus algorithm with 3x throughput improvement.

? Co-authored 4 peer-reviewed conference publications in distributed systems.

SELECTED PEER-REVIEWED JOURNAL & CONFERENCE PUBLICATIONS

[1] E. Rostova, D. Gifford. "Aether: Low-Latency Consensus for Geo-Replicated Datastores." In Proceedings of the USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2023.

[2] E. Rostova, J. Chen, A. Miller. "Byzantine Fault Tolerance Under Dynamic Network Partitions." IEEE Transactions on Parallel and Distributed Systems (TPDS), Vol. 33, No. 8, pp. 1920-1934, 2022.

[3] E. Rostova, R. Morris. "Accelerating Distributed Gradient Descent via Speculative Synchronization." In ACM Symposium on Principles of Distributed Computing (PODC), 2021.

PUBLICATIONS (CONTINUED)

[4] E. Rostova, M. Kaashoek. "Verifiable State Machine Replication with Minimal Overhead." In International Conference on Distributed Computing Systems (ICDCS), 2020.

[5] S. Gupta, E. Rostova, B. Liskov. "Deterministic Transaction Scheduling in Sharded Datastores." ACM SIGMOD Record, Vol. 48, No. 2, 2019.

TEACHING EXPERIENCE

Teaching Assistant · CS 244B: Distributed Systems
Stanford University

Spring 2024

Teaching Assistant · 6.824: Distributed Systems
MIT Department of Electrical Engineering and Computer Science

Fall 2021, Fall 2022

GRANTS, HONORS & AWARDS

- National Science Foundation (NSF) Graduate Research Fellowship, 2018 - 2021
- MIT Presidential Fellowship, 2018
- Carnegie Mellon Senior Leadership Award in Computer Science, 2018
- Best Paper Award, IEEE International Conference on Cloud Computing, 2022

SKILLS & EXPERTISE

Programming: C, C++, Go, Rust, Python, Java, SQL, LaTeX

Systems: Raft, Paxos, Kubernetes, Docker, gRPC, Linux Kernel, Redis, RocksDB